PHASE 1: OFFLINE TRAINING

Training Split D_train (80%)

Standardized Stratified Split
PaySim / IEEE-CIS / ULB

Stage 1: XGBoost Core

Cost-Weighted Ensemble f_core(x)
Regularized Loss (spw=10.0)

Deploy f_core

PHASE 2: VALIDATION & CALIBRATION

Validation Split D_val (10%)

Score Probabilities: p_val = f_core(x)
Held-Out Tuning Partition

Stage 2: Optimal Threshold τ*

100-step Grid Search τ ∈ [0.01, 0.99]
PaySim: τ* = 0.2179 | IEEE: 0.3763

p >= τ*

k=5 OOF Cross-Fitting

Candidate Alerts D_cand (p >= τ*)
Local TreeSHAP ϕ(x) + Train RF

OOF Risk q

Amount-Tiered Safety Calibration

Stratify Alerts into K=3 Tertiles
Calibrate Tier Guards τ_meta,k (ΔFN = 0)

PHASE 3: STREAMING INFERENCE WITH SAFETY GUARD

Live Transaction

Streaming Vector x ∈ R^d
Feature Extraction
(0.002 ms)

Stage 1 Core

Score f_core(x) -> p
Latency: 0.55 ms
Gate: p >= τ* ?

p < τ*

Auto-Cleared Legit

Direct Approval (Bulk Traffic)

p >= τ*

Stage 3 TreeSHAP

Local Attribution ϕ(x)
Exact Explainer
Latency: 3.49 ms

Meta-Classifier

Joint Rep [x, ϕ(x)]
RF Ensemble -> q
Latency: 4.95 ms

Tier Safety Gate

Tier k(Amount)
q >= τ_meta,k ?

Auto-Cleared False Alarm (s = 1)

Escalated Fraud Review (s = 0)